# Module 5: UX and Design Collaboration

Understand UX craft well enough to review designs, run usability tests, and work productively with product designers.



AUTHOR

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Built from the 55-lecture course study guide and extended to the 12-module Product Core curriculum.
Hypothetical examples are labelled; no real company results are invented.

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MODULE 5

# UX and Design Collaboration

Understand UX craft well enough to review designs, run usability tests, and work productively with product designers.

≈ 10 h study

**About the screens in this module**

All wireframes are **illustrative, simplified sketches made for teaching.** They are *not* screenshots of any real company's interface and must never be presented as such.

## 5.1 UX versus UI

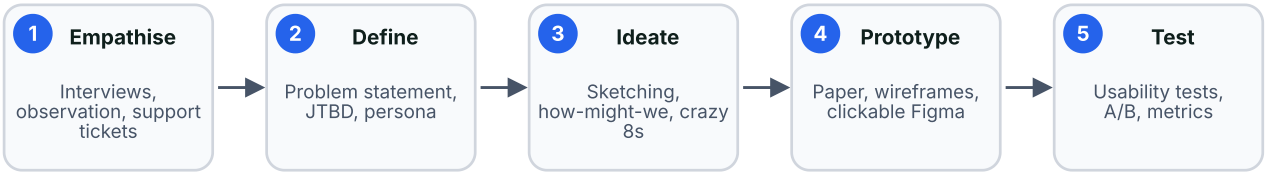
**From your course notes (Lecture 16)**

**UI** is the interface people see and use: text, controls, layout, colour, typography. **UX** is the complete experience of achieving a goal, including usability, accessibility, waiting, trust, service and what happens when something goes wrong. *A large readable Pay button is UI. A checkout that clearly shows the total, works with a screen reader and explains a failed payment is part of UX.*

| | UX (experience) | UI (interface) |
|---|---|---|
| Scope | The whole journey, before/during/after the screen | Visual and interactive layer |
| Questions | Can people achieve their goal? Do they trust it? | Is it clear, consistent, accessible, attractive? |
| Artefacts | Research, journeys, flows, prototypes, test findings | Style guides, components, high-fidelity screens |
| Failure looks like | Users abandon, get lost or distrust | Inconsistent, hard to read or hard to tap |

## 5.2 Design Thinking and Human-Centred Design

**Design Thinking** is a human-centred, iterative approach to problem solving with five stages (Stanford d.school): **Empathise → Define → Ideate → Prototype → Test. Human-Centred Design (HCD)**, standardised in ISO 9241-210, emphasises understanding users, tasks and environments; involving users throughout; evaluating with users; iterating; considering the whole user experience; and using multidisciplinary skills.




**Design Thinking stages mapped to PM activities.** Stages loop; testing often sends you back to Define or Ideate. The double-diamond view (discover, define, develop, deliver) shows the same idea as diverge–converge twice — see the Framework Library.

> **From your course notes (Lecture 16)**
>
> A possible design sequence is: define the problem and requirements, map the flow, sketch or wireframe, build a prototype, test, revise and implement — *iterative, not mandatory.* Include users with varied needs and abilities. A **wireframe** shows structure; a **prototype** lets people experience or test interactions.

## 5.3 Nielsen's 10 usability heuristics

Jakob Nielsen's ten general principles for interaction design are a fast way to review any screen (a **heuristic evaluation**).

| # | Heuristic | Meaning | Example (checkout) |
|---|---|---|---|
| 1 | **Visibility of system status** | Keep users informed with timely feedback | “Processing payment…” spinner with a timeout message |
| 2 | **Match between system and real world** | Speak the user's language | “Delivery fee”, not “ancillary charge code 4” |
| 3 | **User control and freedom** | Easy undo/exit | “Edit cart” and “Cancel” always available |
| 4 | **Consistency and standards** | Follow conventions | Same “Pay” button position on every step |
| 5 | **Error prevention** | Prevent problems before they occur | Disable “Pay” until a payment method is chosen; confirm address |
| 6 | **Recognition rather than recall** | Make options visible | Show saved cards, not “enter the last four digits” |
| 7 | **Flexibility and efficiency of use** | Shortcuts for experts | “Order again”, saved addresses |
| 8 | **Aesthetic and minimalist design** | Remove irrelevant info | Hide secondary promotions during payment |
| 9 | **Help users recognise, diagnose, recover from errors** | Plain-language errors with a fix | “Your card was declined. Try another card or UPI.” |
| 10 | **Help and documentation** | Searchable, task-focused help | Link to refund FAQ from the order screen |

> **From your course notes (Companion ch. 3; Picture lesson)**
>
> Review **visibility of status, clear labels, error recovery, keyboard and screen-reader access, colour contrast and understandable prices.** *A design review (heuristic evaluation) can find possible problems but does not replace testing with real users.* An AI review is a useful draft, not a substitute.

**How to run a heuristic evaluation:** 3–5 evaluators review independently against the list, rate each problem's **severity** (0 cosmetic … 4 catastrophe), then consolidate. It is cheap and fast but misses problems real users hit.

# 5.4 Laws of UX that every PM should know

## 5.4.1 Jakob's Law

Users spend most of their time on *other* sites/apps, so they expect yours to work the same way. **Implication:** follow conventions (cart icon top-right, logo links home) unless you have a strong, tested reason. Innovate on value, not on basic mechanics.

## 5.4.2 Hick's Law

The time to choose increases with the number and complexity of choices.

```
Hick-Hyman law:  T = b × log₂(n + 1)
T = decision time; n = number of equally likely choices; b = empirically
fitted constant
```

**Illustrative calculation (b = 0.1 s, a made-up constant for illustration)**

7 menu options: T = 0.1 × $\log_2(8)$ = 0.1 × 3 = **0.30 s**. 15 options: T = 0.1 × $\log_2(16)$ = 0.1 × 4 = **0.40 s**. Doubling the options adds one bit, not double the time — but real interfaces also add scanning and comprehension costs. **Implication:** group, prioritise, progressively disclose; don't hide key actions.

## 5.4.3 Fitts's Law

The time to reach a target depends on distance and size: closer and larger targets are faster to hit.

```
Fitts's law (Shannon form):  T = a + b × log₂(D / W + 1)
D = distance to target; W = target width (along the movement axis); a, b =
constants
Index of difficulty ID = log₂(D / W + 1)
```

**Worked example**

A button 40 px wide at a distance of 400 px: ID = $\log_2(400/40 + 1)$ = $\log_2(11)$ = **3.46 bits**. Making it 80 px wide: ID = $\log_2(6)$ = **2.58 bits** — about **25% lower**. For touch, thumb reach zones and minimum target sizes (≈ 44×44 pt iOS guidance; 48×48 dp Android/Material guidance; WCAG 2.2 requires at least 24×24 CSS px as a minimum) matter more than the formula. **Implication:** make primary actions large and in easy reach; keep destructive actions small and apart.

### 5.4.4 Gestalt principles

How people perceive visual groups:

| Principle | Idea | Use |
|---|---|---|
| Proximity | Close items are grouped | Space between form sections |
| Similarity | Similar items are related | Same style for all primary buttons |
| Common region | Items in a shared boundary are grouped | Card around the order summary |
| Continuity | Eyes follow smooth lines | Aligned lists and stepper progress |
| Closure | We complete incomplete shapes | Minimal icons |
| Figure–ground | We separate focus from background | Modal overlays dim the page |
| Focal point | Distinct items draw attention | One highlighted primary CTA |

### 5.4.5 Cognitive Load Theory

Working memory is limited. **Intrinsic load** is the inherent difficulty of the task; **extraneous load** is added by poor design (clutter, confusing labels); **germane load** is effort that builds understanding. Designers reduce extraneous load: chunk information, use progressive disclosure, defaults, recognisable patterns and consistent layouts. Miller's “7±2” is often over-quoted — treat it as a reminder that capacity is small, not as a design rule.

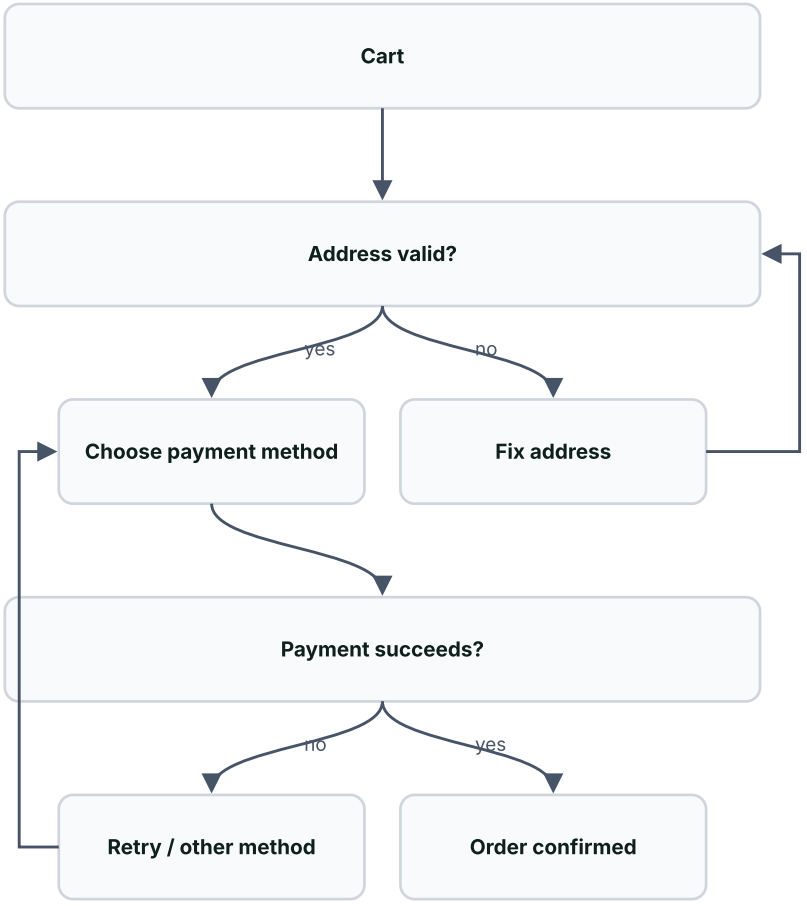
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## 5.5 Information architecture, user flows and task flows

- **Information architecture (IA)** organises content so people can find it: structure, **navigation**, **labelling**, **search**. Methods: **card sorting** (how users group items), **tree testing** (can users find items in your proposed structure?).
- A **user flow** maps paths and decisions through the product toward a goal; a **task flow** is one task in detail, typically for a single user type.

**From your course notes (Lecture 5; Companion ch. 3)**

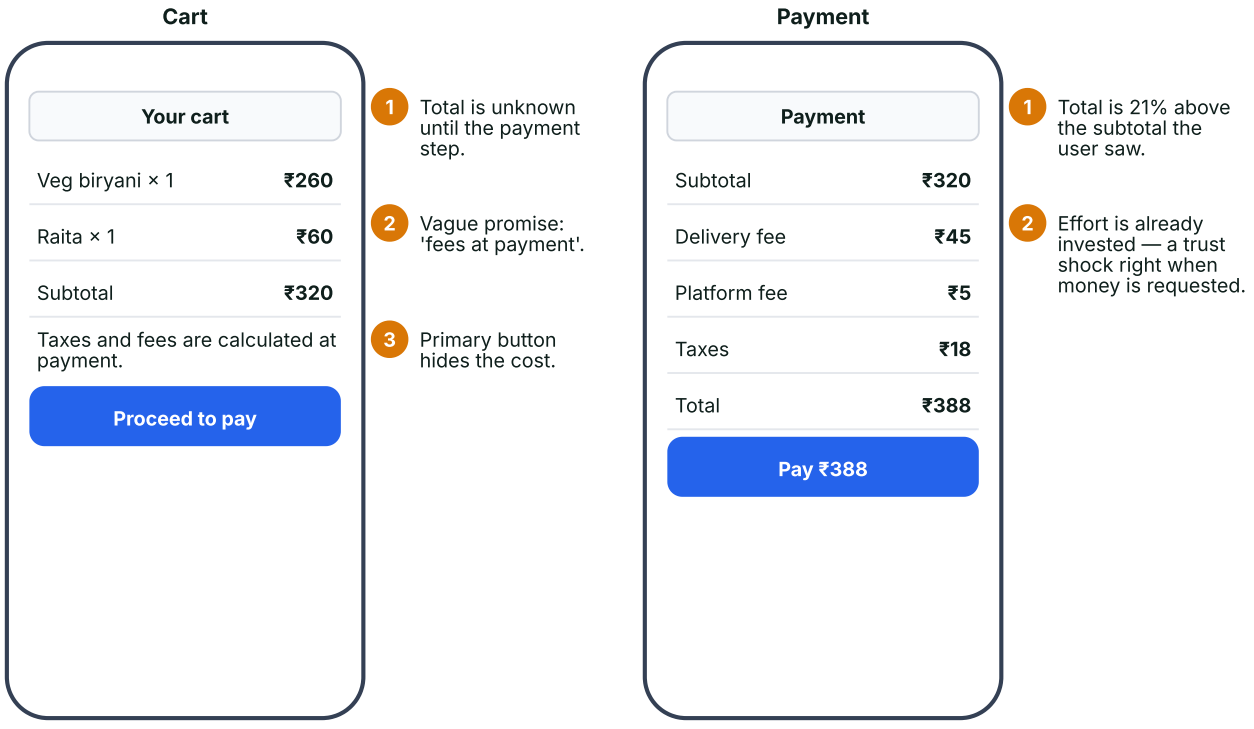
A journey can contain many flows. Include branches: valid address, invalid address, payment failure, retry and cancellation — and label the desired outcome for each. Start/end markers and connectors make the sequence readable. Use the appropriate level of detail; *a functional flow, not backend architecture.*



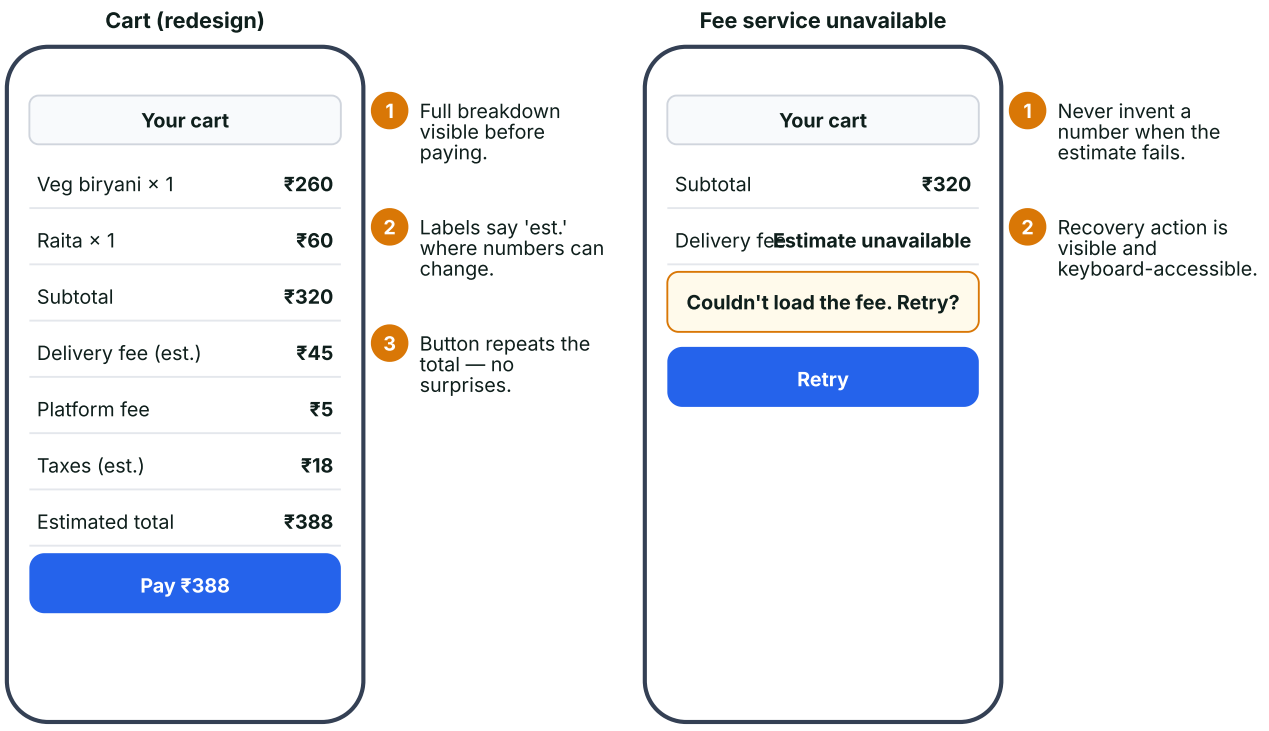

**Task flow — pay for an order (hypothetical).** Decision points and recovery loops are where bugs and abandonment hide. Every “no” branch needs a designed state.

# 5.6 Wireframes and prototypes: low to high fidelity

| Fidelity | What it is | Use for | Risk |
|---|---|---|---|
| **Sketch / paper** | Hand-drawn boxes | Exploring many ideas in minutes | Hard to share remotely |
| **Low-fi wireframe** | Grey boxes, real labels, no visual polish | Layout, flow, content priority | People critique visuals instead |
| **High-fi mockup** | Real visual design | Brand, UI details, stakeholder review | Looks “finished”; invites premature sign-off |
| **Interactive prototype** | Linked screens with transitions | Usability tests, stakeholder demos | Not connected to a real backend; may mislead about feasibility |




**Before — checkout with late fees (illustrative wireframe, not a real product screenshot).** Hypothetical baseline. The total (₹388) is 21% higher than the subtotal shown earlier (388 ÷ 320 = 1.21). This is the pattern interviewees in the earlier modules described.




**After — fee shown early, with honest fallback (illustrative wireframe, not a real product screenshot).** Hypothetical redesign. It follows heuristics 1 (visibility), 5 (error prevention), 9 (error recovery). Hypothesis to test: earlier fee display raises checkout completion without raising complaints (Module 7).

**From your course notes (Companion ch. 17; Lecture 16)**

"Show an estimated total before checkout." In scope: item cost and an estimate based on the address. Out of scope: changing the fee policy. Acceptance: **label estimates, refresh when the address changes, and show a clear fallback when the fee service fails.** Ask design about *empty, loading, error, keyboard and accessible states.* A prototype is not necessarily connected to a real backend.

## 5.6.1 Clickable prototype example

A clickable prototype for the redesign would link: *Cart (redesign)* → *Pay* → *Confirmation*, plus a branch *Cart (fee unavailable)* → *Retry* → *Cart (redesign)*. In Figma: duplicate frames, name them, switch to **Prototype** mode, drag a connection from the button to the next frame, set the trigger (On click) and animation (Instant/Smart animate), then present and share a view-only link. Test it with 5 participants using a task such as: "Order this meal and tell me the total you expect to pay *before* you tap pay."

## 5.7 Design systems, responsive design and mobile vs web

- A **design system** is a shared library of design tokens (colours, spacing, type), reusable **components**, patterns and guidelines — plus code. Benefits: consistency, speed, accessibility baked in. As a PM: *don't request one-off components without a reason; use existing ones; contribute needs via the system's process.*
- **Responsive design** adapts layouts to screen sizes (breakpoints; fluid grids; flexible images). Design mobile-first, then enhance.
- **Mobile vs web UX:**

| | **Mobile app** | **Responsive web** |
|---|---|---|
| Interaction | Touch, gestures, thumb reach; smaller screens | Mouse/keyboard and touch |
| Capabilities | Camera, push notifications, biometrics | Limited by browser; no install friction |
| Constraints | App-store review, OS versions | Cross-browser testing, performance on slow networks |
| Design focus | One main task per screen, bottom navigation | Wider layouts, persistent navigation |

> **From your course notes (Companion ch. 17; Extra topic 14)**
>
> Avoid changing a shared production library without permission. Review variations, reusable components and responsive layouts with your designer; compare decisions to acceptance criteria and record unresolved questions in the PRD.

## 5.8 Accessibility

Accessibility means people with different abilities can perceive, understand and use the product. **WCAG** (Web Content Accessibility Guidelines, currently 2.2) is organised around four principles — **Perceivable, Operable, Understandable, Robust** — with conformance levels A, AA, AAA (AA is the common target).

**From your course notes (Extra topic 14)**

Inclusive design also considers varied devices, language, connectivity, literacy and circumstances. **Accessibility is an ongoing product quality concern, not a finishing decoration.** A form should use labels in addition to colour, have visible keyboard focus, describe errors in text and work with assistive technology; **a delivery estimate must not rely only on a red or green indicator.** Involve people with disabilities; use automated checks *alongside* manual and assistive-technology testing; ask qualified specialists about applicable standards and legal obligations.

**Contrast ratio** is $(L_1 + 0.05) \div (L_2 + 0.05)$, where L is relative luminance (lighter ÷ darker). WCAG AA requires **4.5:1** for normal text and **3:1** for large text (≥ 24 px, or ≥ 18.66 px bold) and for UI component boundaries.

**Worked check — contrast of common text colours on white (computed)**

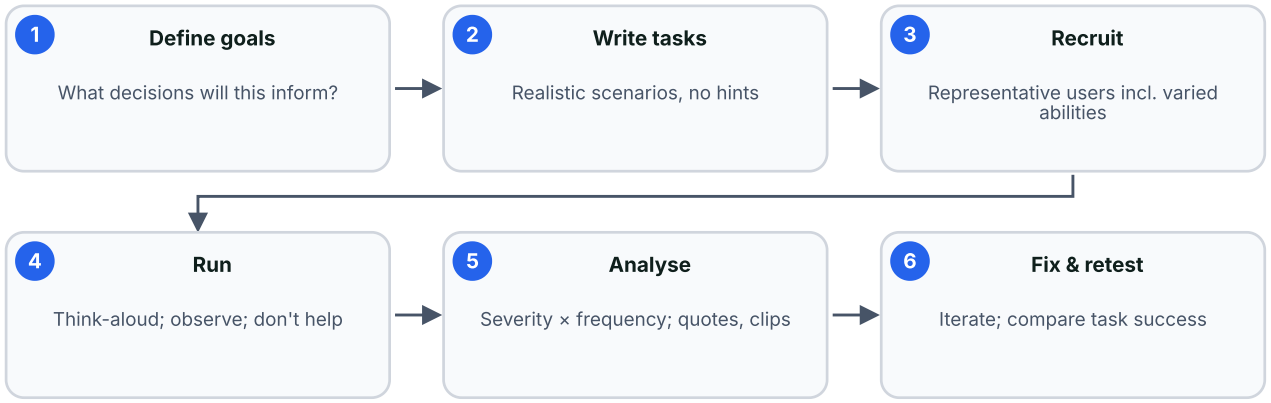
| Text colour | Ratio on white | AA normal text (4.5:1)? |
|---|---|---|
| #101828 (navy) | 17.75 : 1 | Pass |
| #2563EB (royal blue) | 5.17 : 1 | Pass |
| #667085 (grey) | 4.97 : 1 | Pass |
| #767676 | 4.54 : 1 | Pass (barely) |
| #777777 | 4.48 : 1 | **Fail** |
| #949494 | 3.03 : 1 | **Fail** (large text/UI only) |
| #10B981 (emerald) | 2.54 : 1 | **Fail** — white text on this green is also 2.54 : 1 |

This is why this handbook uses a deeper green (#047857) for filled buttons with white text: the brand emerald alone does not meet AA for text.

**PM accessibility checklist:** semantic structure and labels · keyboard operability and visible focus · contrast · text alternatives for images · captions for video · target sizes · error messages in text with guidance · no information by colour alone · respects reduced motion · tested with a screen reader on key flows · accessibility acceptance criteria in every story.

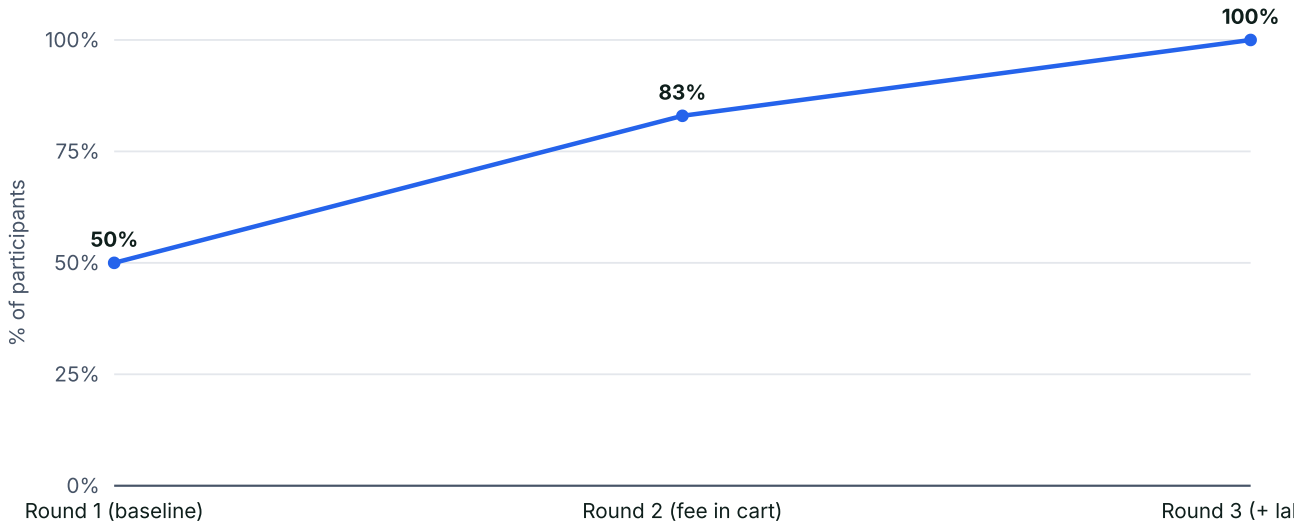
# 5.9 Usability testing

A **usability test** observes representative users attempting realistic tasks. Typical: **5–8 participants per round** — Nielsen and Landauer's model estimates one user finds ~31% of problems, so five users find about $1 - (1 - 0.31)^5 \approx$ **84%**, which is why iterating with small rounds beats one big test (the figure is a model for finding *problems*, not for estimating *rates*).




**Running a moderated usability test.** Always get consent, protect privacy and use a prototype that matches the question. Use test findings to improve the design, then retest.

**Task wording:** bad — “Click the fee breakdown.” Good — “You're about to pay for this meal. How much will you be charged in total? Tell me what you're looking at.” **Measures:** task success, time on task, errors, confidence, plus **SUS** (System Usability Scale, 10 questions; average across products ≈ 68 on a 0–100 scale).




**Task success across design iterations (hypothetical, 6 participants per round).** Hypothetical. With 6 participants each step is one or two people — treat the shape as a direction, then confirm with live behavioural data. Measure whether the *real* completion rate improves.

**From your course notes (Lecture 16; Picture lesson)**

**Observe real people completing a task.** Make a prototype, test with users with varied needs, improve and retest. The session contrasts *discovering a problem* with *testing whether people can navigate a proposed design.* A design review can find possible problems but does not replace testing with real users.

## 5.10 A/B testing in UX

A/B tests measure **behaviour at scale** (completion rates, clicks to goal), not understanding. Use usability tests to learn *why*, A/B tests to learn *how much*. Avoid testing tiny cosmetic differences with small traffic; test meaningful hypotheses and pair with guardrails (complaints, errors). See Module 7 for the statistics.

## 5.11 Design critique and working with product designers

**Design critique** is structured feedback on a design against goals — not personal taste. Good format: *goal and constraints → what's being reviewed → feedback framed as observations tied to user/goal ("I notice... it may cause...") → questions → decisions.* Avoid solutioneering in comments.

**From your course notes (Lecture 16)**

The instructor described modifying a designer's file without coordination, causing inconsistencies and confusion. **Agree on ownership, explain user and business context, use the design system and resolve disagreements through evidence.** *The claim that PMs should never edit Figma and that UI is solely a developer's responsibility is too rigid; roles depend on the team.* Designers contribute to solutions; PMs can learn and create designs when appropriate.

**How to be a great partner to designers**

1. Bring the **problem and evidence**, not a solution. 2. Involve them in **discovery** from day one. 3. Share constraints early (tech, legal, timing). 4. Give **specific, user-centred feedback**. 5. Protect time for exploration and for **accessibility and edge states**. 6. Respect craft: ask “what are you optimising for?”. 7. Decide *together* what to test.

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# 5.12 Figma fundamentals for PMs

**From your course notes (Companion ch. 17)**

Figma Design is a collaborative design tool. A **frame** represents a screen or container; text, shapes and components form the interface; a **component** is a reusable element; a **prototype** connects screens to illustrate a flow. Editing rights and features depend on the team and plan. Beginner exercise: create a personal file, add a phone-sized frame, place a heading, meal card, price and button; duplicate the frame and change it to show the total fee; in prototype mode connect the button to the next frame, preview and test it. PM workflow: use **comments** to describe the user problem and evidence (“Three test participants missed the total; could we make it visible before payment?”). Share prototype links with appropriate permissions.

| Figma concept | What it is | PM relevance |
|---|---|---|
| Frame | Screen/container | Check screens match the flow |
| Auto layout | Responsive spacing rules | Explains how content grows (long names, translations) |
| Component & variants | Reusable element with states (default, hover, error, disabled) | Ensures states are designed |
| Styles / variables | Colour, type, spacing tokens | Consistency, theming |
| Prototype | Clickable flow | Testing and demos |
| Comments | Contextual feedback | Give specific, evidence-based feedback |
| Dev Mode | Specs and code snippets for engineers | Handoff clarity |

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## 5.13 Journey mapping, onboarding and checkout optimisation

- **Onboarding** — get users to their first meaningful value quickly: reduce steps, defer non-essential fields, use progressive profiling, show progress, explain permissions *when needed* (just-in-time), offer a sample/empty-state starter.
- **Checkout** — show the total early, minimise fields, offer guest checkout and saved payment methods, give clear error recovery, keep the primary action consistent.

| Friction | UX fix | Metric to watch | Guardrail |
|---|---|---|---|
| Long sign-up form | Ask only essentials; social/phone sign-in where appropriate | Sign-up completion | Fraud/abuse rate |
| Permission prompt on first open | Ask in context, explain value | Permission grant rate | Uninstall rate |
| Hidden fees | Show total early with a breakdown | Checkout completion | Complaints |
| Payment failure dead end | Retry, alternative methods, saved cart | Payment success after failure | Duplicate-charge reports |
| Empty first screen | Sample content, guided first action | Activation within 7 days | Support contacts |

**Common onboarding and checkout friction → fix → metric.** Every fix is a hypothesis. Test it, and watch the guardrail so you do not trade trust for conversion.

## 5.14 Error states and empty states

- **Error states:** say what happened, why (if useful), and how to fix it; preserve user input; offer a way forward; log for diagnosis. Avoid blame (“Invalid input!”) and codes alone.
- **Empty states:** teach the product — explain the benefit, show the primary action, optionally sample data.
- **Loading states:** skeleton screens beat endless spinners; give time estimates for long tasks.

| State | Weak | Strong |
|---|---|---|
| Payment error | “Error 502” | “We couldn't reach your bank. Your card hasn't been charged. Try again or use UPI.” |
| Empty orders | Blank | “No orders yet. Your first order will appear here. Browse popular meals →” |
| Offline | Frozen app | “You're offline. Showing saved items; ordering resumes when you're back online.” |

## 5.15 UX metrics

**Google's HEART framework** (Happiness, Engagement, Adoption, Retention, Task success) with the **Goals–Signals–Metrics** process helps choose UX metrics.

| Dimension | Goal | Signal | Metric |
|---|---|---|---|
| Happiness | Users feel confident paying | Satisfaction after order | CSAT; checkout SUS |
| Engagement | Users interact with the breakdown | Taps on fee details | % sessions opening breakdown |
| Adoption | First-time buyers complete checkout | First order placed | First-order conversion |
| Retention | Buyers come back | Second order in 14 days | 14-day repeat rate |
| Task success | People pay without errors | Completed vs failed attempts | Completion rate; error rate; time on task |

**HEART applied to a checkout redesign (hypothetical).** Pick the few dimensions that match the goal; not every project needs all five. Add guardrails (complaints, accessibility issues).

Other UX measures: **task success rate**, **time on task**, **error rate**, **SUS**, **CSAT** (satisfaction), **CES** (effort), **NPS** (recommendation — widely used, with limitations as a UX measure), **accessibility issue count**, **support contacts per 1,000 orders**.

## 5.16 Design handoff

A good handoff is a **conversation**, not a file drop: walk through the flow, states (empty/loading/error/offline), responsive behaviour, accessibility, copy, analytics events, and edge cases. Provide: final frames, component references, spacing/typography specs (Dev Mode), prototype link, acceptance criteria, **open questions**, and an agreed **QA review** (design reviews the build before release).

**Handoff checklist:** all states designed · copy final (or flagged) · accessibility requirements listed · events named · animation specs · assets exported · decisions logged · QA owner named · definition of done includes design review.

## 5.17 Exercises with model answers

**Exercise 1 · Heuristic evaluation**

Evaluate the “Before” checkout screens against the 10 heuristics. List three violations with severity.

**Model answer**

(1) **Visibility of system status** — the final cost isn't visible until payment (severity 3: causes abandonment). (2) **Match with the real world / consistency** — “fees are calculated at payment” hides a concept users need now (severity 2). (3) **Error prevention** — no way to preview the total before committing (severity 3). Also note heuristic 9: no designed failure state if fee calculation fails. Recommend the “After” design and test it.

**Exercise 2 · Apply Fitts's law**

A primary button is 48 px wide and 500 px from the thumb's resting point. What is the index of difficulty? What happens if you widen it to 96 px?

**Model answer**

ID = $\log_2(500/48 + 1) = \log_2(11.42) \approx$ **3.51 bits**. With 96 px: $\log_2(500/96 + 1) = \log_2(6.21) \approx$ **2.64 bits** — about 25% easier. In practice also move it into the thumb zone and keep a ≥ 44 px height.

**Exercise 3 · Accessibility check**

Your error text is #949494 on white, 14 px. What is wrong and how do you fix it?

**Model answer**

Contrast is ~3.03:1; normal text needs ≥ 4.5:1. Fix: darken to ≈ #767676 (4.54:1) or darker (e.g., #595959 = 7.0:1); don't rely on colour alone — add an icon and the text “Error:”. Test with a screen reader that the error is announced and associated with the field.

**Exercise 4 · Plan a test**

You redesigned onboarding. Write three realistic tasks and two metrics for a usability test with 6 users.

**Model answer**

Tasks: (1) “You've just downloaded the app. Set it up so it can deliver to your office.” (2) “Find a vegetarian meal under ₹300 and add it to the cart.” (3) “Check what you'd pay in total.” Metrics: task success rate, time on task, errors/hesitations, confidence rating (1–5), SUS at the end. Pre-define success (e.g., ≥ 5 of 6 complete task 3 unaided).

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## 5.18 Interview perspective

**“How would you improve the checkout experience of a food-delivery app?” — structure**

1) Clarify goal (completion? AOV? trust?) and users. 2) Map the journey and find the biggest friction with data (funnel by step/device) and research. 3) Hypotheses: fees shown late, payment failure recovery, form length. 4) Solutions with trade-offs (show fee early; guest checkout; retry). 5) Prioritise (impact/effort/risk). 6) Design considerations: states, accessibility, honest fallbacks. 7) Test: usability test then A/B with guardrails. *Model answer highlights:* “I'd start with the funnel to find where users leave, then interview people who abandoned. If fees surprise them, I'd test showing a full breakdown in the cart with 'estimate' labels and a clear fallback. Primary metric: checkout completion; guardrails: complaints and payment errors; segment by new vs returning users.” *Mistakes:* jumping to UI changes without evidence; ignoring failure states and accessibility.

**“How do you work with designers when you disagree?” — 60-second answer**

“I start by getting aligned on the user problem and the success metric. Then I ask what the designer is optimising for and share the evidence I have. If we still disagree, we agree on a quick test — a usability session or A/B — and a decision rule in advance. I give feedback as observations tied to the user, not preferences, and make sure the final decision is documented.”

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## 5.19 Key takeaways

- **UX** = whole experience; **UI** = the interface layer.

- Use **heuristics** and the **laws of UX** (Jakob, Hick, Fitts), **Gestalt** and **cognitive load** to review designs quickly.
- Design **all states**: empty, loading, error, offline, success.
- **Accessibility** is a quality requirement: check contrast, keyboard, labels, target sizes.
- **Usability tests with ~5–8 users per round** find most problems; A/B tests measure behaviour at scale.
- Partner with designers by bringing **problems and evidence**, respecting ownership and the design system.
- Make handoff a **conversation** with states, events, accessibility and QA agreed.

# 5.20 Check your understanding

CHECK YOUR UNDERSTANDING

**Which statement correctly contrasts UX and UI?**

A. UI is the whole experience; UX is the buttons.

B. UX is the whole experience; UI is the interface people see and use.

C. They are the same.

D. UX applies only to mobile apps.

**Why:** UX covers the full experience of achieving a goal, including usability, trust and failure handling; UI is the visual and interactive layer.

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**Hick's law says decision time...**

A. Decreases with more options

B. Increases with the number and complexity of choices

C. Is constant

D. Depends only on colour

**Why:** More (and more complex) choices take longer to process; group and prioritise options.

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**Fitts's law implies a primary mobile button should be...**

A. Small and far from the thumb

B. Large and within easy reach

C. Hidden in a menu

D. Red

**Why:** Time to reach a target falls as it gets closer and larger.

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**Normal body text needs at least what contrast ratio for WCAG AA?**

A. 2 : 1

B. 3 : 1

C. 4.5 : 1

D. 21 : 1

**Why:** WCAG 2.2 AA requires 4.5:1 for normal text, 3:1 for large text and UI components.

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**Why run several small usability rounds rather than one large one?**

A. It is cheaper per participant only

B. Each round finds most problems, you fix them, then retest to find the next layer

C. Statistics require it

D. Designers prefer it

**Why:** Small samples find most usability problems; iterating compounds the benefit.

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**A good error message...**

A. Shows an error code only

B. Blames the user

C. Explains what happened and how to recover, in plain language

D. Is hidden

**Why:** Heuristic 9: help users recognise, diagnose and recover from errors.

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## 5.21 Recommended resources

- Nielsen Norman Group: 10 usability heuristics and heuristic evaluations.
- Laws of UX — concise write-ups of Jakob's, Hick's, Fitts's laws and more.
- W3C: WCAG 2 overview and How to Meet WCAG (Quick Reference).
- Figma Help: Share files and prototypes.
- Google HEART framework (Rodden, Hutchinson, Fu).
- Stanford d.school: Design Thinking Bootleg.